Athinoula A. Martinos Center For Biomedical Imaging

Global Biohealth Engineering Hackathon

Open-source Pulseq programming with self-supervised AI image reconstruction

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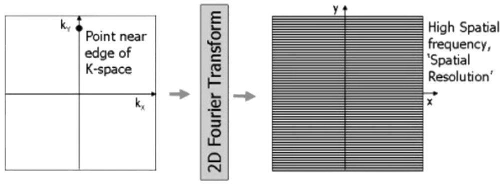
Motivation

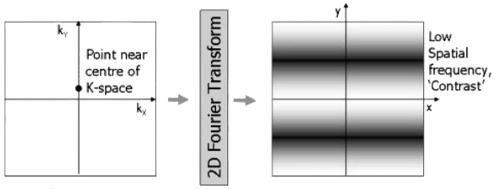
Balancing Scan Time and Image Quality in Accelerated MRI



Longer Scan

- More k-space data
- Higher image quality
- Fewer artifacts
- Greater patient motion risk

Scan Time–Image Quality Trade-off



Shorter Scan

- Less k-space data
- Lower image quality
- More aliasing artifacts
- Lower patient motion risk

Our goal is to reduce scan time
while preserving image quality through advanced reconstruction.

02 Motivation

Experimental Workflow

Overall Framework

Sequences

- 9 Pulseq sequence files
- Fully sampled reference
- Acceleration: R = 2–8
- Uniform / random Cartesian

→

Acquisition

- Multi-coil k-space Raw
- scanner data (.dat)

→

Reconstruction

- zero-filled
- SENSE
- L1 Wavelet
- AI reconstruction (SSL)

→

Evaluation

- Relative error
- PSNR
  (Peak Signal-to-Noise Ratio)
- SSIM
  (Structural Similarity Index Measure)
- Error maps

↓

Results & Comparison

01 Pulseq
Pulse Sequence Design

Pulseq Framework

- Pulseq is an open-source, vendor-independent framework for MRI pulse sequence programming.
- It generates standardized sequence files by defining RF pulses, gradients, and ADC events, enabling flexible sequence design and rapid prototyping.
- Its hardware-independent design enables the same pulse sequence to be executed across different MRI scanner vendors with minimal modification.

Fixed Acquisition Parameters

In this study, Pulseq was used to generate MRI pulse sequences with different acceleration factors while keeping all other acquisition parameters fixed.

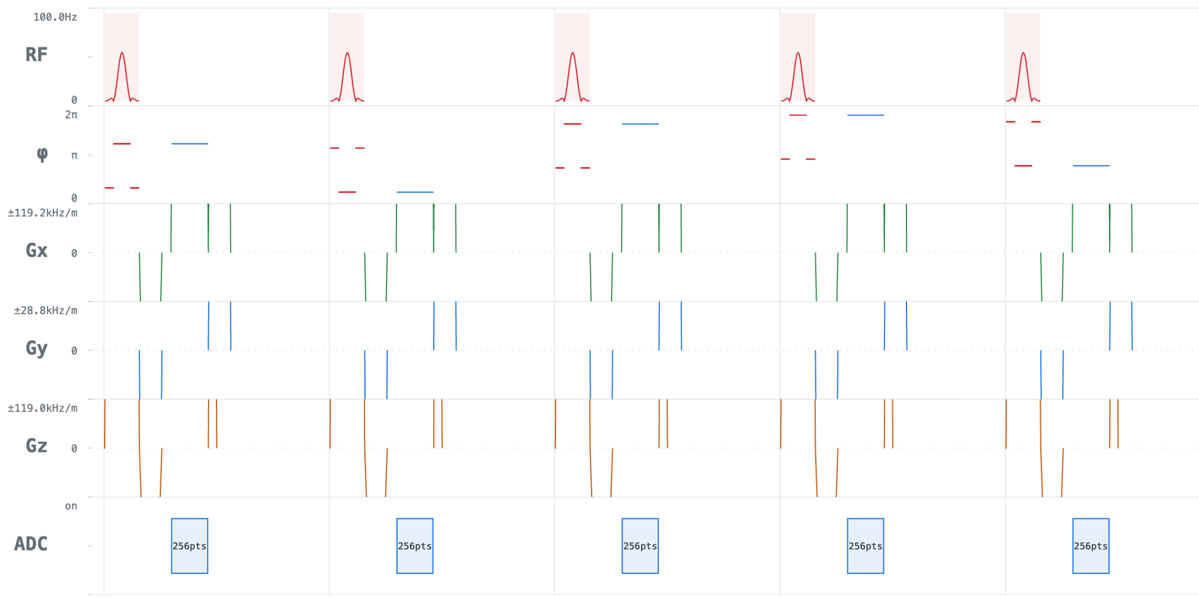
Settings	FOV	Matrix	Pixel Size	Slice	Flip Angle	TE / TR	RF Pulse	Readout Flat
Value	240 × 240 mm	256 × 256	0.94 mm	6 mm × 1	*TE: 15°	6.0 / 20.0 ms	Sinc, 3.0 ms	3.2 ms

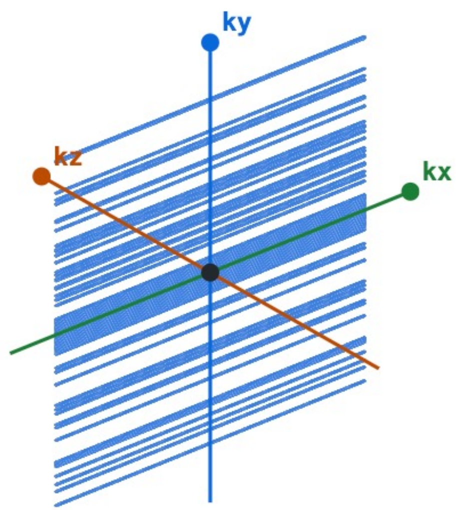
*TE: Time from RF excitation to echo acquisition
*TR: Time between successive RF excitations.

Pulseq Acquisition Parameters by Acceleration Factor

	R = 1 (full)	R = 2	R = 4	R = 6	R = 8
Acquired k_y lines	256 / 256	128	64	43	32
ACS	—	24	24	24	24
Effective R	1.00	2.00	4.00	5.95	8.00
ADC samples	65,536	32,768	16,384	11,008	8,192
Total scan time	7.12 s	4.56 s	3.28 s	2.86 s	2.64 s

Example GRE Timing Diagram & k-space

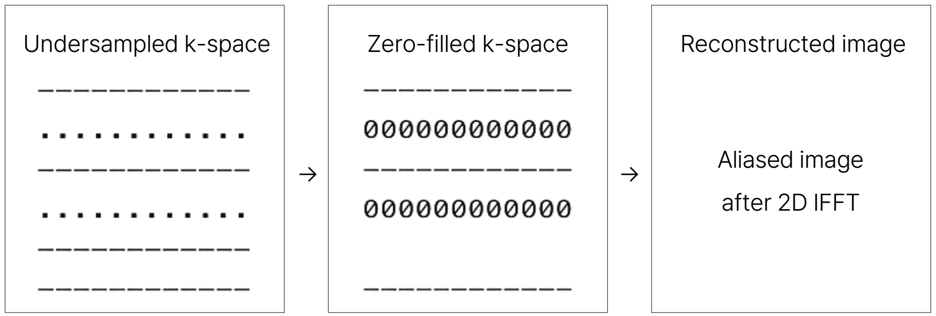




01 Reconstruction

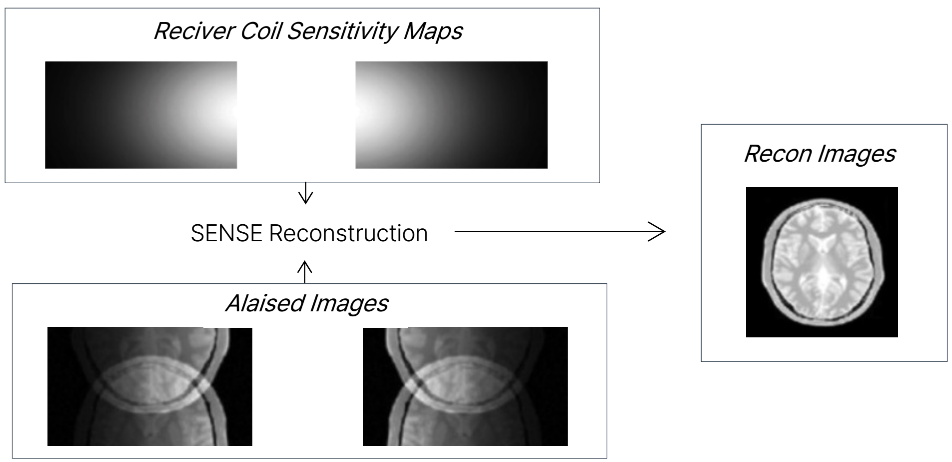
Reconstruction Methods

Strategy 01 Zero-filled



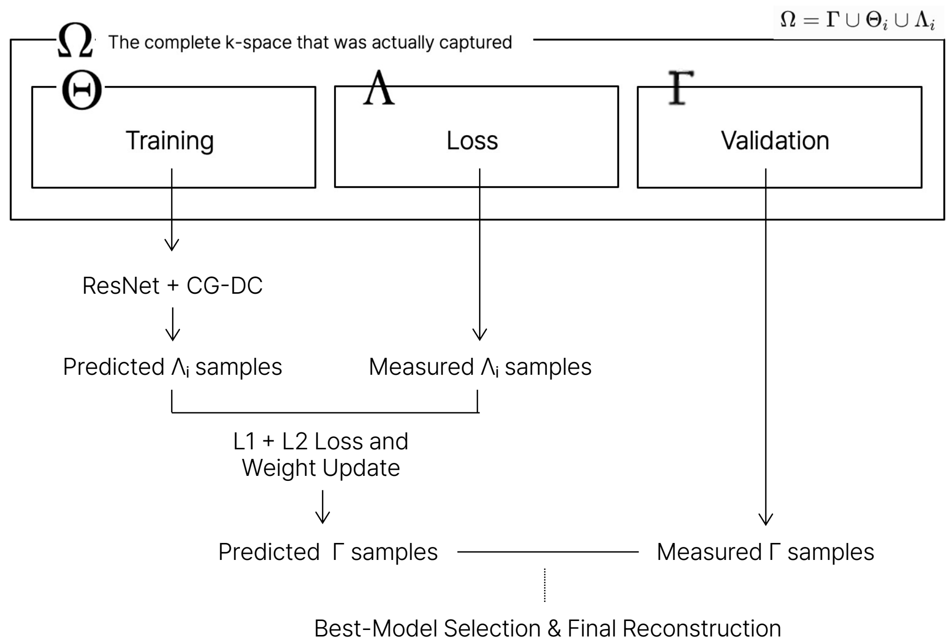
- Fills the missing k-space locations with zeros.
- Then applies the inverse Fourier transform directly.

Strategy 02 SENSE



Source: Khalil et al., Magn. Reson. Imaging, 80 (2021), 58–70.

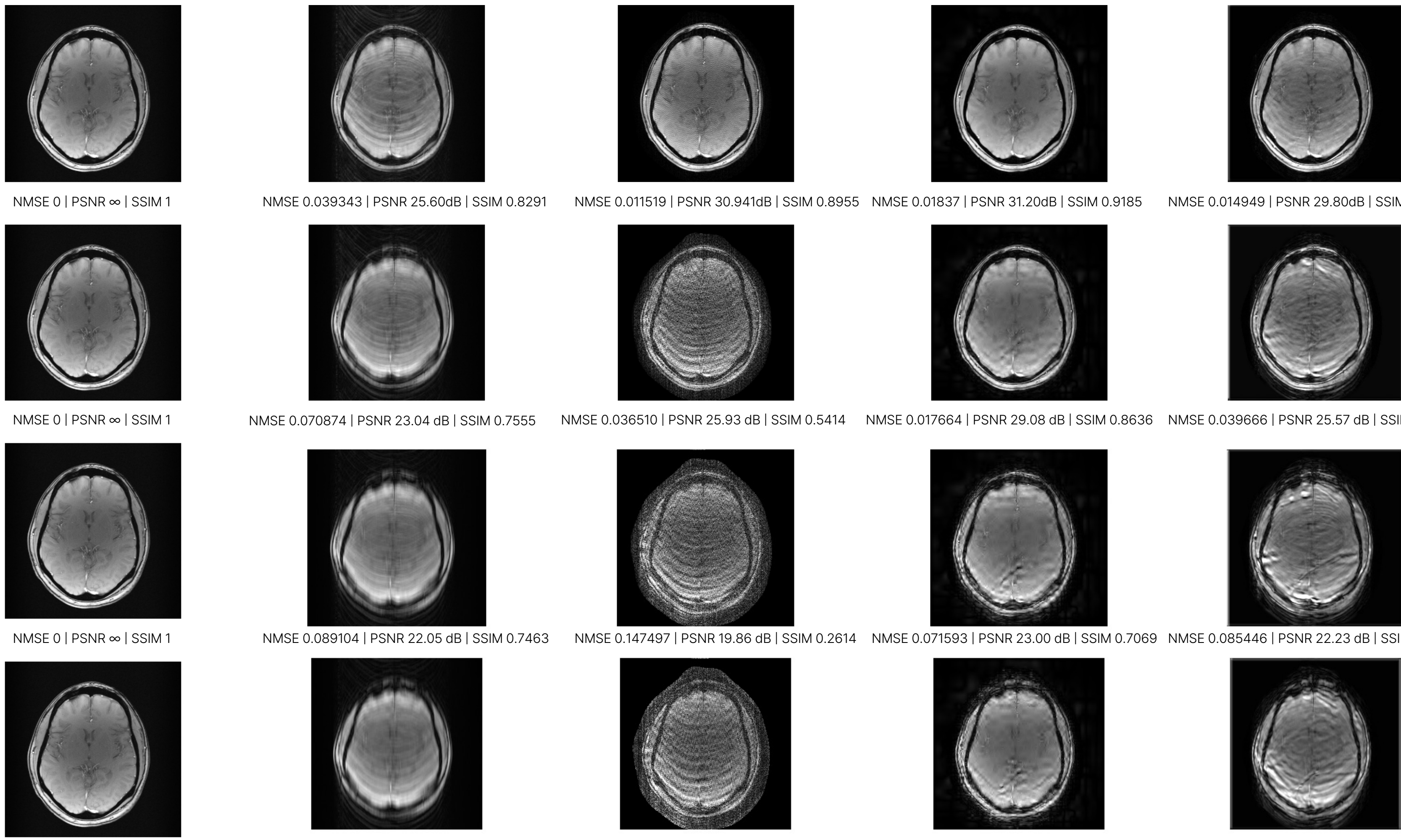
Strategy 03 self-supervised learning



- In clinical MRI, fully sampled data are difficult and slow to acquire.
- SSL hides part of the undersampled k-space and uses it as a target.
- The model learns to recover missing information directly from the acquired data, without a fully sampled reference.

Image Comparison

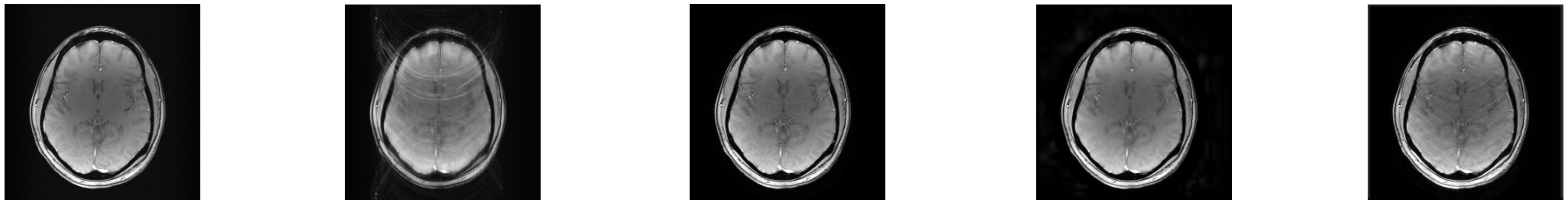
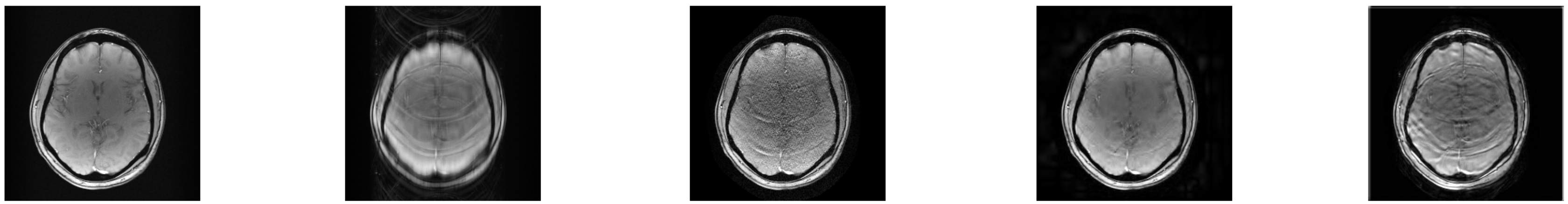
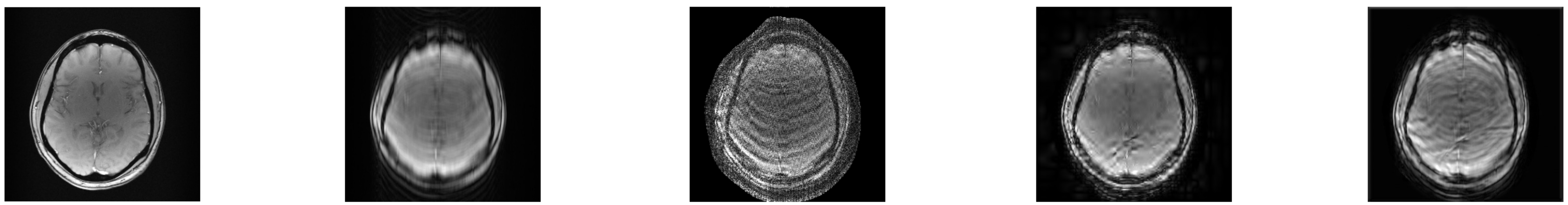
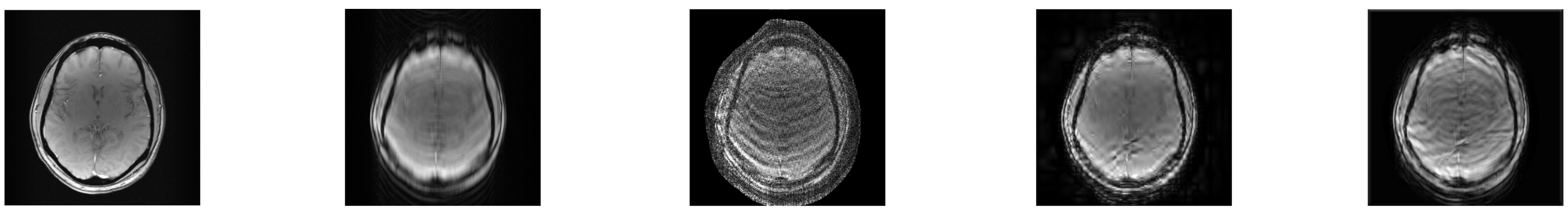
Random Cartesian Undersampling



	Reference (R=1)	Zero Filled	SENSE	L1 Wavelet	AI Recon
R = 2	NMSE 0 | PSNR ∞ | SSIM 1	NMSE 0.039343 | PSNR 25.60dB | SSIM 0.8291	NMSE 0.011519 | PSNR 30.941dB | SSIM 0.8955	NMSE 0.01837 | PSNR 31.20dB | SSIM 0.9185	NMSE 0.014949 | PSNR 29.80dB | SSIM 0.8834
R = 4	NMSE 0 | PSNR ∞ | SSIM 1	NMSE 0.070874 | PSNR 23.04 dB | SSIM 0.7555	NMSE 0.036510 | PSNR 25.93 dB | SSIM 0.5414	NMSE 0.017664 | PSNR 29.08 dB | SSIM 0.8636	NMSE 0.039666 | PSNR 25.57 dB | SSIM 0.6859
R = 6	NMSE 0 | PSNR ∞ | SSIM 1	NMSE 0.089104 | PSNR 22.05 dB | SSIM 0.7463	NMSE 0.147497 | PSNR 19.86 dB | SSIM 0.2614	NMSE 0.071593 | PSNR 23.00 dB | SSIM 0.7069	NMSE 0.085446 | PSNR 22.23 dB | SSIM 0.6349
R = 8	NMSE 0 | PSNR ∞ | SSIM 1	NMSE 0.008307 | PSNR 32.36dB | SSIM 0.9280	NMSE 0.008307 | PSNR 32.36dB | SSIM 0.9280	NMSE 0.008307 | PSNR 32.36dB | SSIM 0.9280	NMSE 0.008307 | PSNR 32.36dB | SSIM 0.9280

02 Results
Image Comparison

Uniform Cartesian Undersampling

	Reference (R=1)	Zero Filled	SENSE	L1 Wavelet	AI Recon
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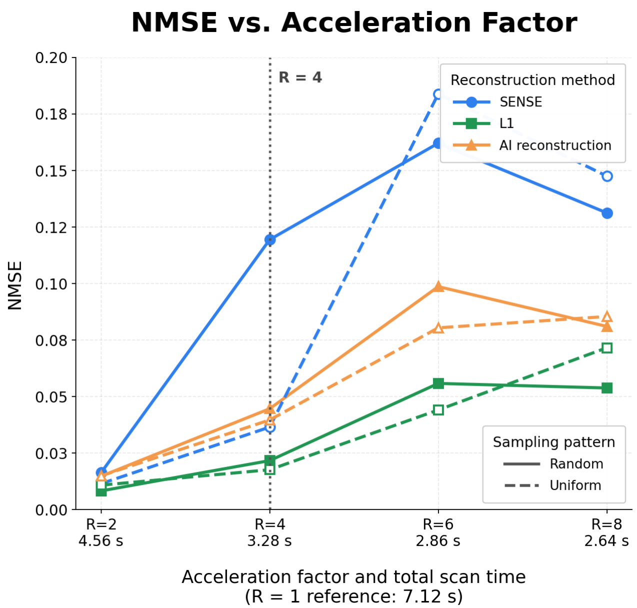
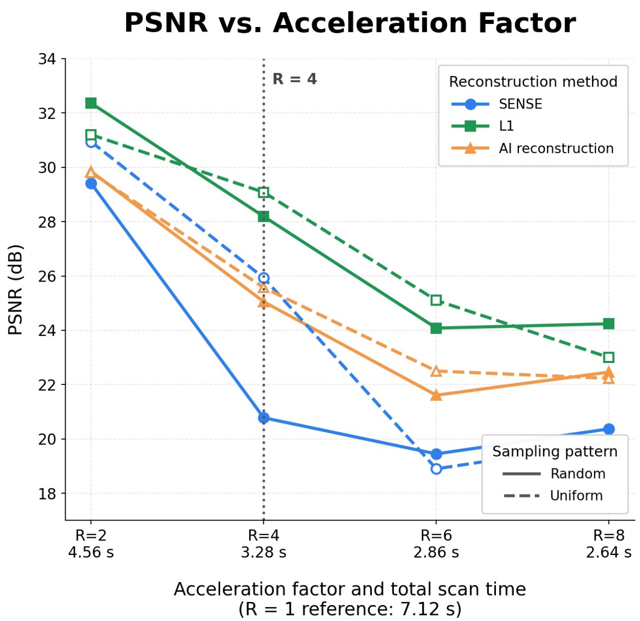
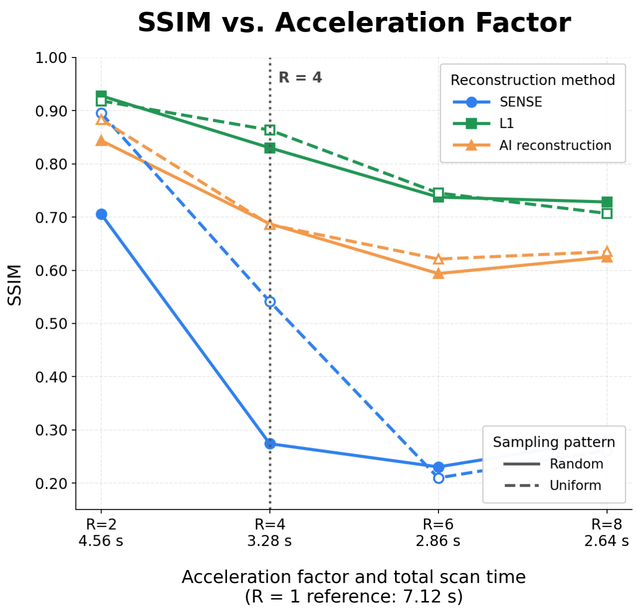
Performance Across Acceleration Factors







- ✓ R = 4 reduced the scan time from 7.12 s to 3.28 s while maintaining relatively good reconstruction quality.
- ✓ Beyond R = 4, the additional time saving was small, but NMSE increased and PSNR and SSIM decreased clearly.
- ✓ Therefore, R = 4 provides the most practical balance between scan-time reduction and image quality

06 Results Discussion

Key findings

- ✓ Increasing the acceleration factor reduced the scan time, but also decreased the reconstruction quality.

- ✓ The scan time decreased from 7.12 s at R = 1 to 3.28 s at R = 4, while the image quality was still acceptable.

  After R = 4, the additional time saving was small, but NMSE, PSNR, SSIM, and residual images became clearly worse.

  Therefore, R = 4 was identified as the practical sweet spot between scan speed and image quality.

- ✓ AI reconstruction was competitive at some settings but was not consistently superior.

  This suggests that the learned model did not consistently outperform physics-based methods because its relatively shallow network had limited capacity to model complex aliasing patterns, especially under limited training data and insufficient hyperparameter tuning.

- ✓ L1 showed the most stable performance across random and uniform sampling.

  Its sparsity constraint likely reduced sensitivity to the exact aliasing pattern, while SENSE depended more strongly on sampling geometry and coil sensitivity.